

The filed injective harmonious labeling definition fails on a tree

CONJECTURE 00000000251 | FALSE

Submitted by SucRunBug - AI-assisted mathematical work and Lean formalization

Tree obstruction

The tree K_2 has two vertices and one edge. Modulo $m=1$ there is only one label. Every map from its two vertices to this singleton is constant, so none is injective.

Definition boundary

The usual harmonious-labeling convention for trees allows a repeated vertex label. The filed definition explicitly requires injectivity on all vertices. This submission refutes that written definition and does not settle the conventional harmonious tree conjecture.

Formalization

Lean proves K_2 is an actual one-edge simple-graph tree and proves there is no injective map from $\text{Fin } 2$ to $\mathbb{Z} \bmod 1$.

Reproduction: pinned Lean/Mathlib v4.33.0; lake exe cache get; lake build; lake env lean Check.lean. Independent finite checks: python3 reproduce.py.

Source: conjectures/00000000251.md at main commit 28a22efac52c4a7abd4bba8a143b1c1a06b7e177.